

Surya Vardhan Yalavarthi

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PROFESSIONAL SUMMARY

Data Scientist with experience in building analytics frameworks and machine learning models. Proficient in Python, R, SQL, and dashboard delivery, with a track record of actionable insights, improved data accuracy, and support for senior teams.

EDUCATION

University of Cincinnati

Master of Engineering, Computer Science

• GPA: 3.83

Aug 2024 - Apr 2026

Cincinnati, OH

TECHNICAL SKILLS

- **Soft Skills:** Remote Collaboration, Independent Problem Solving, Adaptability
- **Data Engineering:** SQL, Pandas, NumPy, ETL Pipelines, Data Extraction, Unstructured Data Processing
- **NLP & Agents:** Natural Language Processing (NLTK), Conversational AI, Text Analysis, Recommender Systems
- **Data Visualization:** Power BI, Matplotlib, Seaborn
- **Core ML/AI:** Python, Scikit-Learn, XGBoost, PyTorch, R Programming, Statistical Modeling, Deep Learning (CNNs)
- **Other Skills:** Java, JavaScript, Docker, Git, C++

EXPERIENCE

Community Dreams Foundation

Data Analyst

Dec 2025 - Present

Sebring, FL

- Collaborated on a multi-phase personal finance analytics initiative, owning data modeling, metric definition, simulation analysis, and dashboard delivery for household financial wellness use cases.
- Explored structured and unstructured synthetic transaction and debt datasets to uncover trends, patterns, and actionable financial insights including cash-flow trends, expense structure, savings behavior, and debt burden under multiple economic scenarios.
- Generated analytical models and performed scenario evaluation (income shocks, expense spikes) using Python and SQL, supporting debt payoff strategy comparisons for stakeholders.
- Refined dashboards and data visualizations using Python libraries and Power BI, translating complex outputs into stakeholder-friendly insights.

Andhra Pradesh State FiberNet Ltd

Machine Learning Intern

Jul 2023 - Oct 2023

Vijayawada, India

- Developed a personalized recommender system in Python that analyzed user behavior and increased relevant content suggestions, boosting user engagement by 15%.
- Built, evaluated, and optimized predictive models in Python using machine learning techniques and real-world datasets.
- Researched and applied model evaluation techniques and presented clear summaries to senior leadership, directly supporting their decisions on technology selection for key projects.
- Cleaned and organized large datasets independently using Python pandas, which improved data accuracy and cut preprocessing time by 30% for downstream analysis.

PROJECTS

Household Finance Wellness & Debt Optimization Lab

2025 - Present

- Built a personal finance analytics framework using Python and SQL to analyze household transactions, savings, and debt data.
- Implemented spend classification (needs vs wants), financial wellness scoring, and debt payoff simulations (snowball and avalanche).
- Developed dashboards and visual reports to present educational, non-advisory financial insights.

Brain Tumor Detection using CNN

Dec 2022 - Apr 2023

- Implemented CNN-based image classification achieving 96% accuracy in detecting anomalies from MRI scans.
- Executed rigorous data validation steps, including Data Augmentation, to ensure model reliability.
- Documented data preprocessing steps clearly to allow for reproducibility and process understanding.

Body Weight Prediction using 2D Images

Dec 2023 - Apr 2024

- Built a deep neural network integrated with XGBoost to predict body weight from 2D images.
- Conducted independent research on feature selection techniques, summarizing results for model optimization.

Data-Driven Recommendation System for User Engagement

Jul 2023 - Nov 2023

- Created a data-driven recommender system using Python and analytics to enhance user engagement through personalized interactions.

CERTIFICATIONS

- Microsoft Azure Fundamentals (AZ-900)
- Oracle Cloud Infrastructure 2023 Certified Architect Associate
- Google TensorFlow Developer Certificate
- Oracle – Cloud Infrastructure 2023 Certified Data Science Professional
- ChatGPT Prompt Engineering for Developers